

7 (a) In the photoelectric effect experiment, explain what is meant by

(i) work function energy of a metal,

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..... [1]

(ii) the stopping potential.

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..... [1]

(b) To demonstrate the photoelectric phenomena, a setup consisting of two narrow strips of aluminium foil attached to an aluminium can as shown in Fig. 7.1 is constructed.

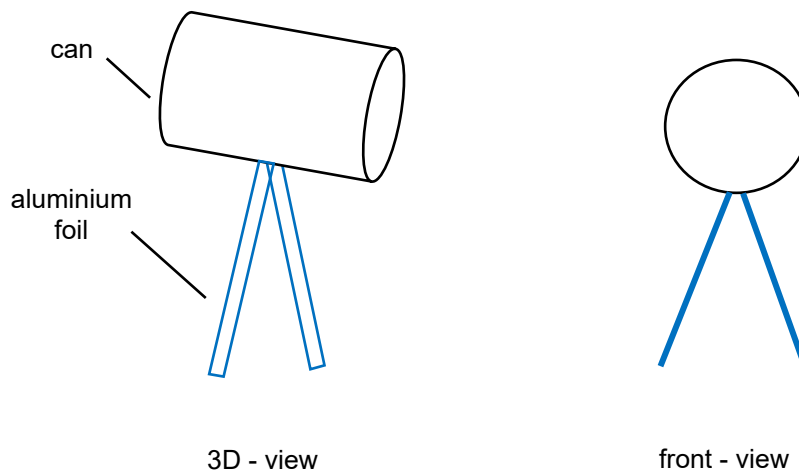


Fig. 7.1

The can and strips are **positively charged** and the two strips repel each other.

(i) During the setting up of the apparatus, sand paper is used to grind the surface portion of the aluminium can where light is incident on.

Suggest a reason for doing this.

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..... [1]

(ii) The following changes are made independently. Describe and explain, in each case, the effect on the two strip of aluminium foil, if any.

1. Ultra-violet light, whose frequency is higher than the threshold frequency of aluminium, is incident on the aluminium can.

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 [1]

2. A radioactive source that undergoes negative beta decay is placed near the aluminium can.

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 [2]

3. Red light, whose frequency is lower than the threshold frequency of aluminium, is incident on the aluminium can. The intensity of the red light is gradually increased.

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 [2]

(iii) Explain how the observation in **(b)(ii)(3)** gives evidences for the quantum behavior of light.

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 [3]

